ClipSAT : Egyptian Scholastic Test: EST II Mathematics Level 2 [ANSWER KEY]

Unit 1.01: Complex Numbers, Modulus & Polar Form

Q1. If i = sqrt{-1} , which of the following is equivalent to the complex expression frac{7 + 4i}{2 - i} ?

[x] A: 2 + 3i

[ ] B: 2 - 3i

[ ] C: frac{10 + 15i}{3}

[ ] D: frac{18 + 15i}{5}

Solution: To simplify the quotient of two complex numbers, multiply both the numerator and the denominator by the complex conjugate of the denominator, which is 2 + i : frac{7 + 4i}{2 - i} = frac{7 + 4i2 + i}{2 - i2 + i} Expand the numerator using the distributive property FOIL: 7 + 4i2 + i = 14 + 7i + 8i + 4i^2 Since i^2 = -1 : 14 + 15i + 4-1 = 14 - 4 + 15i = 10 + 15i Expand the denominator using the difference of squares: 2 - i2 + i = 2^2 - i^2 = 4 - -1 = 5 Now, divide each term in the numerator by the denominator: frac{10 + 15i}{5} = frac{10}{5} + frac{15i}{5} = 2 + 3i Common Misconceptions: - Choice B occurs from an arithmetic sign error in the imaginary term. - Choice C occurs if the student mistakenly calculates the denominator as 2^2 - 1^2 = 3 by forgetting that -i^2 = --1 = +1 . - Choice D occurs if the student takes i^2 = 1 , yielding 14 + 4 = 18 in the numerator.

Q2. Consider the complex number z = 3 + 4i . What is the magnitude of z ?

[x] A: 5

[ ] B: 7

[ ] C: 12

[ ] D: 25

Solution: The magnitude of a complex number z = a + bi is given by |z| = sqrt{a^2 + b^2} . For z = 3 + 4i , we have a = 3 and b = 4 . Therefore, the magnitude is |z| = sqrt{3^2 + 4^2} = sqrt{9 + 16} = sqrt{25} = 5 .

Q3. Let the complex number w be defined as w = a+bi + 3-2i, where a and b are real numbers. Given that |w|^2 = 25 and w lies in the first quadrant, find the value of a+b.

[ ] A: 5

[x] B: 6

[ ] C: 7

[ ] D: 8

Solution: Let w = a+bi + 3-2i = a+3 + b-2i. We are given that |w|^2 = 25, so a+3^2 + b-2^2 = 25. Since w lies in the first quadrant, a+3 > 0 and b-2 > 0. We need to find integer or simple fractional solutions for a+3 and b-2 that sum of squares is 25. The possible pairs of squares are 9, 16 and 16, 9. Case 1: a+3^2 = 9 and b-2^2 = 16. This gives a+3 = pm 3 and b-2 = pm 4. Since a+3 > 0 and b-2 > 0, we have a+3 = 3 and b-2 = 4. This leads to a=0 and b=6. However, if a=0, then w = 3 + 4i, and a+b = 0+6 = 6. Case 2: a+3^2 = 16 and b-2^2 = 9. This gives a+3 = pm 4 and b-2 = pm 3. Since a+3 > 0 and b-2 > 0, we have a+3 = 4 and b-2 = 3. This leads to a=1 and b=5. In this case, w = 4 + 3i, which is in the first quadrant. We want to find a+b. So, a+b = 1+5 = 6. Both cases yield a+b=6.

Q4. Let z_1 = 2 + 3i and z_2 = 1 - i. Consider the complex number w such that w = frac{z_1^2}{z_2} . If w = a + bi, where a and b are real numbers, find the value of b.

[ ] A: frac{17}{2}

[x] B: -frac{17}{2}

[ ] C: frac{11}{2}

[ ] D: -frac{11}{2}

Solution: First, calculate z_1^2: z_1^2 = 2 + 3i^2 = 2^2 + 223i + 3i^2 = 4 + 12i + 9i^2 = 4 + 12i - 9 = -5 + 12i . Next, divide z_1^2 by z_2: w = frac{-5 + 12i}{1 - i} . To simplify this division, multiply the numerator and the denominator by the conjugate of the denominator, which is 1 + i: w = frac{-5 + 12i}{1 - i} times frac{1 + i}{1 + i} = frac{-5 + 12i1 + i}{1 - i1 + i} . The numerator is: -51 + -5i + 12i1 + 12ii = -5 - 5i + 12i + 12i^2 = -5 + 7i - 12 = -17 + 7i . The denominator is: 1^2 - i^2 = 1 - -1 = 1 + 1 = 2 . So, w = frac{-17 + 7i}{2} = -frac{17}{2} + frac{7}{2}i . Given that w = a + bi, we have a = -frac{17}{2} and b = frac{7}{2}. The question asks for the value of b. Therefore, b = frac{7}{2}.

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Unit 1.01: Complex Numbers, Modulus & Polar Form

Q5. Let z = 3 - 4i. What is the real part of z?

[x] A: 3
[ ] B: -4
[ ] C: 5
[ ] D: 7

Solution: A complex number is generally expressed in the form a + bi, where a is the real part and b is the imaginary part. In the given complex number z = 3 - 4i, the real part is the term that does not have i multiplied by it. Therefore, the real part of z is 3.

Q6. Let z_1 and z_2 be the roots of the quadratic equation x^2 - 4x + 5 = 0 . Consider a complex number w such that w = z_1^2 + z_2^2. If w is expressed in the form a + bi, where a and b are real numbers, find the value of a - b.

[x] A: 6
[ ] B: 8
[ ] C: 10
[ ] D: 12

Solution: The quadratic equation is x^2 - 4x + 5 = 0 . By Vieta's formulas, the sum of the roots is z_1 + z_2 = --4/1 = 4 and the product of the roots is z_1 z_2 = 5/1 = 5 . We are interested in w = z_1^2 + z_2^2. We can rewrite this as w = z_1 + z_2^2 - 2z_1 z_2. Substituting the values from Vieta's formulas: w = 4^2 - 25 w = 16 - 10 w = 6. So, w = 6. When expressed in the form a + bi, we have a = 6 and b = 0. The question asks for a - b. a - b = 6 - 0 = 6.

Q7. Consider the complex number z = 3 + 4i . What is the modulus of z ?

[x] A: 5
[ ] B: 7
[ ] C: 12
[ ] D: 25

Solution: The modulus of a complex number z = a + bi is given by |z| = sqrt{a^2 + b^2} . For z = 3 + 4i , we have a = 3 and b = 4 . Therefore, the modulus is |z| = sqrt{3^2 + 4^2} = sqrt{9 + 16} = sqrt{25} = 5 .

Q8. What is i^2 ?

[x] A: -1
[ ] B: 1
[ ] C: i
[ ] D: -i

Solution: Applying Complex Numbers principles, By definition i^2=-1 . Therefore, the verified correct option is -1.